

Play and Resistance: Intersecting Identities and Implicit Biases in Gamified Educational Tools

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ABSTRACT

As Human-Computer Interaction (HCI) researchers, we can create digital tools which facilitate growth, liberate, and heal, or create tools for trauma, discrimination, and increasing social power divides. Thus, it is extremely important to follow intersectionally grounded, qualitatively sound approaches with a diversity of participants situated across different axes of oppression, not only to improve the effectiveness of our explicit interfaces, but also to change the implicit messages and power structures we encode in them. The outcome of this paper, which represents a middle step between several larger studies, is twofold: a definition of the research methodology used to more accurately study bias in gamification in education, and outreach to potential study participants for the followup study. From current literature on the effects of (mis)representation, inaccessible and harmful designs, and oppressive technologies in HCI, we see that a crucial step in HCI research around gamified education tools is missing: the experiences, stories, and design input of marginalised students.

CCS CONCEPTS

• **Social and professional topics** → **User characteristics**; • **Applied computing** → **Education**; • **Human-centered computing** → **Human computer interaction (HCI)**;

KEYWORDS

gamification, educational tools, intersectionality, representation, norms, biases, narratives

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1 INTRODUCTION

The purpose of this paper is to identify implicit biases in existing gamified educational tools as an attempt to reduce inequitable barriers to education. This work continues our evaluation of gamification

usage in the higher education context and discriminatory factors in current design standards. Given the nature of the study, described more in section 3, it is vital to use rigorous anti-oppressive, qualitative methods with as diverse group of users as possible and we believe this publication will help us with that. This study's intent is not to remove all bias from the gamified educational systems. This would be impossible. We see the importance of identifying elements of oppression, resistance, play and inequitable students' experiences with respect to the same educational tools.

Digital environments allow teachers to share different knowledge media with their students, iterate with them, and monitor key indicators of performance and growth [30]. These became a necessity during the COVID19 pandemics, accelerating the need for remote learning, and remain a vital part of the hybrid educational process [12]. Digital and hybrid environments fundamentally benefit marginalised groups, allowing remote communities access to top-tier education and disabled people a variety of tools absent from analogue settings [55]. Gamification, which is defined as the use of game elements (e.g. points, badges, meaningful stories, etc) in non-game contexts [14], flawlessly integrates into the concept of hybrid education. Students can work on the gamified assignments from the comforts of their homes; educational system enhancements (e.g. avatars, badges) are digital tools, which are being used more and more [12].

Prior stages of this research project used quantitative studies to evaluate motivation [42] and state of flow [34, 43], across gamified educational tools. Based on literature reviews [21, 25, 51, 52, 60] we can conclude that most of the published papers on the effects of gamification research explicit behavioral and psychological outcomes. For example, the study of the relationship between gamification and learning outcomes [13]. When it comes to biases, there are some papers which discuss the separate aspects of the digital learning environments such as anonymity [33], avatars [6] or gender-based coloring [1, 53]. However, a deeper qualitative evaluation of students' implicit experiences and outcomes, often unrelated to the subject taught by the digital tool itself, is still missing, together with a critical view on the newly emerging educational tools. Researchers agree that digital technologies are more than just a medium for communicating academic material [27], and we are interested in investigating what they are communicating beyond that.

Responding to [10, 16, 46, 61], this study is a crucial step within a larger research project. The far-reaching impacts of who is represented (and who is not) in digital spaces alone shows negative impacts to health, well-being, performance and social cohesion [38]. The very act of who is named or coded (and how) shows similar negative impacts [3]; whether divergent abilities are given leadership in the design of digital tools echoes immediate and long-term harms [56].

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As personalisation of tools is increasingly adopted, understanding the needs of people across identity factors is a critical step toward simultaneously reducing harm and increasing educational experiences and outcomes. Orji's work on personalising strategies for improved health outcomes demonstrates the need for this work with digital tools [35]; in the case of tools benefiting from AI: those harmful biases grow, sometimes exponentially [17]. The negative effects of stereotype threat are well-researched [44]. Since the most recent findings on this topic show the importance of mitigating these stereotypes and biases [53] we see the need for the further research of students' experiences in an increasingly hybridized and gamified classroom. Results may focus on gamification tools, but - hopefully - contribute to interdisciplinary shifting patterns in how we construct digital learning spaces and educational methods altogether. It is those most in need of these changes, the under-represented and actively oppressed populations, who are denied platforms for their voices, needs, and the agency to design better solutions for all.

2 THEORETICAL AND CONCEPTUAL FRAMEWORK

The microcontext for this work is the popularisation of gamification in educational settings. Its macrocontext is the fact that there's a present shared focus on equity and accessibility [15, 57, 58, 63] in ACM research communities¹, which is supported by EU and UK legislation and standards, and the fact that governments are pushing (albeit slowly) toward equal access to and equal rights for education to all citizens.

Maxwell [31], as cited in [48], offers seeing the goals from three perspectives: intellectual, practical and personal. We want to explore the biases which students experience while working with the gamified educational tools, and thus contribute to the body of the research literature on this topic. In practice, that would mean a more refined set of best practices for the educators and developers of such tools. On a personal level, this article contributes to our ongoing research about gamification in education, providing in-depth qualitative research and methods for use cases often restricted to quantitative or non-inclusive approaches.

To situate ourselves, our biases and practices: both authors are white, disabled, of middle class, and reside in Ukraine and Canada. Both authors are non-binary but read as a gender-conforming man and woman, which has resulted in multiple occasions of stereotyping and biases towards both of us in the academic settings in general, and in educational tools in particular. Both authors work extensively within communities for activism and equity in HCI across intersections of identity, and in institutions with students historically marginalised. Their stakes around liberation from ableism, sexism and genderism are rooted in first-hand experiences; their labor within anti-racist, anti-colonial, and anti-classist spaces is not informed by direct experience but through decades of committed grassroots work while living alongside and loving those who experience these oppressions first-hand.

Theoretically, we situate our work in the literature on the existing gamified tools in higher education, discussed in depth in section 3.

We begin our exploration with the following research questions:

- (1) If any, how and where do university students perceive biases (e.g. race, gender, etc) in existing educational gamified tools, and cope with them?
- (2) What do students feel when confronted with elements of bias in educational games?
- (3) Do these experiences of bias relate to the known ones of non-gamified educational systems?

Embedded in these research questions are multiple assumptions about the experiences of the students during the hybrid educational process, including the notion that the narratives can give us valuable insights into lived experiences of the participants. The proposed study uses the conceptual framework developed around three concepts: (1) narrative, (2) self-identity and biases, (3) gamification in higher education (Figure 1).

2.1 Narrative

Narrative research takes as an object of the research the story itself, investigating from the case rather than comparing between cases [50]. Analysing such a story requires a lot of insight into the participant's life, their autobiographical and historical context, and grounded ways of thinking. We will build the context for understanding their experiences based through extended time spent with participants, their self-identity as discussed below. There's always a difference between life as lived and life as told, which involves storytelling – an event between a participant and a researcher. When these stories are being told, the tellers are performing their preferred identities [28]. For this research, we choose to not imply objectivity, but on the contrary - positionality and subjectivity, by taking into account the perspectives of both the participant and the researcher. This approach is inspired by Personal Narrative Group, as described in [50].

2.2 Biases

Bias is a socially constructed, situated psychological phenomenon. As a phenomenon that is inseparable from power dynamics, a purely structuralist narrative approach will not account for the ways that social forms of identity and power affect the experiences elicited. Our presence as researchers affects what participants' disclose, their style of expression, the structure of their story [49]. As stated in [48], researchers are the instruments of the study, hence self-reflexivity will be involved at each step of the project. Genderism, racism, elitism, stereotype threat, pro-/anti-social identity relations, and the other related interpersonal dynamics are not only the biases for study; they are brought into the research setting as well.

2.3 Gamification in higher education

Inspired by Decuyper [12], for the current research we take the stance of a "critical platform gaze" – hence approaching the gamified platforms not as neutral digital tools, but as complex sociotechnical constructs, which are transforming the educational sector on many levels. This project is mainly focused on the existing, customly developed gamified tools for higher education [41, 42]. Digital platforms that serve as intermediates (Khan Academy, Google's

¹Race in HCI collective <https://www.raceinhci.com/>; HCI Guidelines for Gender Equity and Inclusivity <https://www.morgan-klaus.com/gender-guidelines.html>; Queer HCI <https://sites.google.com/view/queer-hci/>

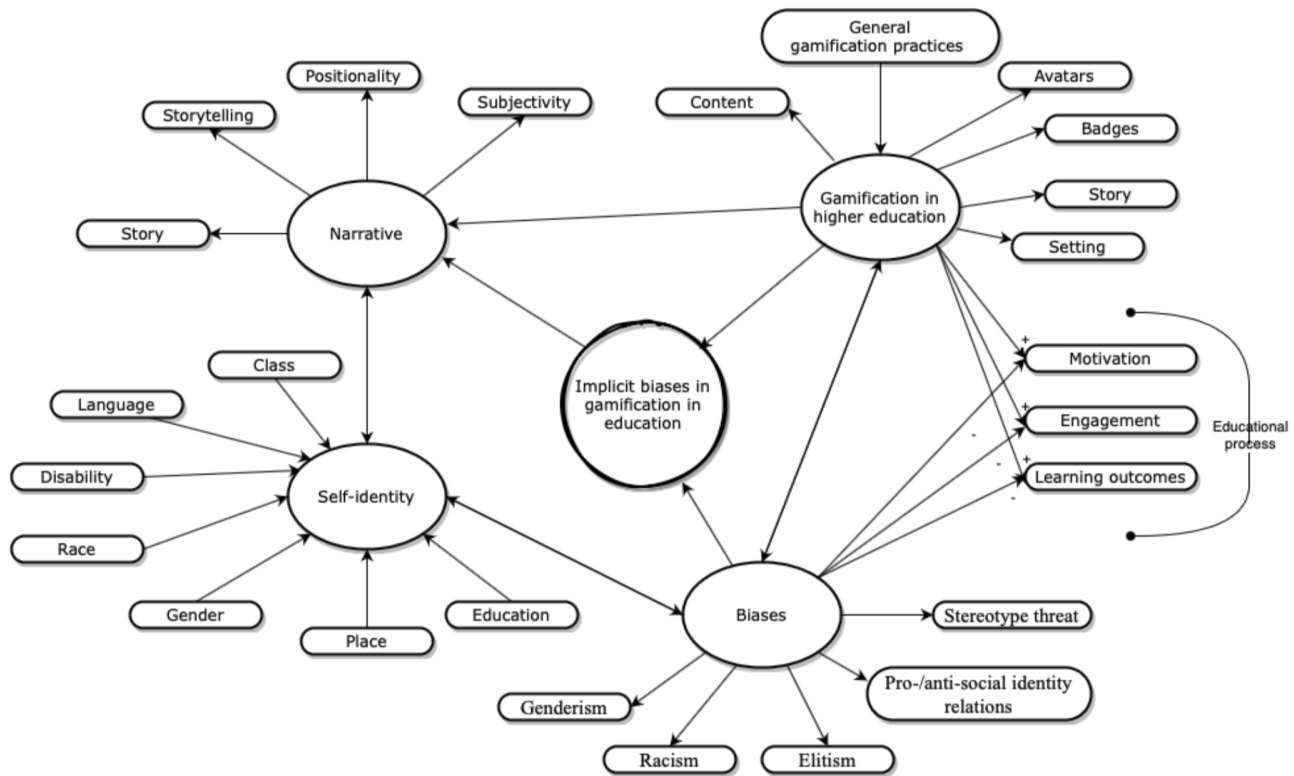


Figure 1: Concepts playing role in the biases which students experience in gamified educational systems

Classroom platform and other popular Learning Management Systems) fall beyond the scope of the current paper. Gamification tools, as digital platforms, are structuring and shaping the attention, decision making, and cognition processes [4]. To make this work as intended, significant attention should be paid to the forms of the tools. For the gamified tools, this could mean investigating what explicit elements (avatars, badges, setting, story, content, etc), implicit messages and with what purpose are being used, how are they shaping the education, and what user identities they generate.

3 RESEARCH APPROACH

We situate our work as narrative research since we seek to investigate the stories of the students about their experiences. To better account for the dynamics discussed above, an intersectionality-sensitive constructionist epistemological framework will be used to inform our narrative theoretical approach.

Since qualitative research is an iterative process, we not only understand that the described plan might change and evolve, we expect and hope it will. Ravitch & Mittenfelner [48] say that research proposals frame future studies, but do not determine them. We want our personal growth and relationship building with communities partnering in this study, our interactions with our peers, mentors, and participants to shape and refine the methods, outcomes, and next steps of this research and the way we discuss it.

Constructionism is particularly sensitive epistemology for this study because it holds that, “all knowledge, and therefore all meaningful reality as such, is contingent upon human practices being constructed in and out of the interaction between human beings and their world, and developed and transmitted within an essentially social context” [11]. Concepts like stigma, oppression, and social power dynamics are social constructs [18]. They result from the privileging of one way of being, thinking, acting, or believing above others. As Bruner [5] describes, everybody tells a story when they speak and they tell these stories differently. In constructionism, our “experiences” are how reality is presented to us; expression is how we describe, frame, and emphasise those experiences. We are given access to experiences as they are expressed by people themselves — as they tell the story of their experience to themselves.

Constructionism and narrative approaches only hold if we can guarantee some degree of shared experience and interpretation of expression [11]. Intersectionality is a framing of identity and experience that takes into account the variety of social identities people are ascribed and how those shape and determine the experiences of folks deemed “non-normative” [8]. People are identified and oppressed across a variety of social signifiers, from skin colour to ability, body shape, and language to age and gender. Intersectionality provides us with a lens to how these identifying elements combine in unique ways; how being a woman and being Black

intersect, for example, is not equal to adding the oppressive experiences of patriarchy and white supremacy. The intersection of these marginalized identities creates unique experiences [7, 8].

The literature on discrimination in gamification is still developing, however, intersectionality was well researched in the game studies. For example, there are studies in the online gaming field [19, 20, 23, 47, 54] that describe the relationship between the benefits, needed coping, and moments of overcoming the harm done by the systemic discrimination. Those papers support the idea that such research requires qualitative methods of self-report; furthermore, health, well-being, coping, power and identity are inseparable [26, 37]. People wish to identify with the representations available to them, evaluating similarities and differences across avatar skin tones, types of dress, facial features, the language used, cultural depictions and so on [24, 38, 39, 62]. Studies show recurring experiences of discrimination and stigma are commonly coped with through normalization [32, 64]. This can mean that, as Passmore & Mandryk [40] suggested, participants are not always willing to disclose painful, stigmatising experiences and even – in some cases – actively suppress their importance [36]. Other papers [19, 57, 59] demonstrate how these considerations apply to dynamics of ability, neurodivergence, and age. Freely eliciting these experiences is by no means easy, however, they are more commonly expressed when there is a trusting relationship with those positioned similarly [2, 20, 47].

3.1 Participant selection and data collection

Invitations to participate will be posted online (at forums, online or hybrid conference spaces, in the relevant Slack channels, and through the personal network) including the goal of the research, short information on the future efforts of the selected participants, motivation to contribute to this research project, and the gratitude for their consideration. With this publication, we hope to be able to reach more folks from the CHI community and involve them in our project as well, and are inviting everyone who is interested in participating to email the authors. Potential participants will complete a brief 1-2 minute survey. Included in this survey is an eligibility prompt: “experience with the gamified educational system in the university in the past year”, and a brief demographics form eliciting general information on participant gender, age, culture, ethnicity, disability, and socioeconomic status. Up to 10 participants will be recruited for a follow-up interview from this pool, exact day and time to be agreed upon with each participant individually. This will allow for a diverse series of intersections across axes of identities. As the study is meant to capture a snapshot of the diversity in stories told around marginalization, bias, and gamified education tools, more detailed interviews are preferable to a larger participant pool.

Data collection will include a series of semi-structured interviews. For the limit of this paper, only the planned in depth interview questions are provided in the Appendix A. The actual interviews will also contain introduction, demographic and concluding sections. A rigidly structured approach would disservice the phenomena of study (bias) as a type of vulnerable experience that participants are prone to guarding [40]. The semi-structured approach will allow for flexibility in establishing rapport, engendering trust, and gently

probing guarded experiences [45]. Interviews will be designed to engender trust, build rapport, and be a safe context to explore potentially harmful past experiences. They will begin with a brief, casual conversation where the study will be explained, that no deception will be employed, and interviewees will have their rights to privacy, consent, and disclosure clearly presented to them. Important aspect of the selected research approach is that not all of what’s being said in the interviews is considered a narrative [50]. Based on the theoretical interest of this research, introductory questions about positionality will contribute to the understanding of the context and not to the narrative itself. However, additional attention will be put to the position in the story participants choose for themselves, and how they are performing their identities: victims of the circumstances, being the ones in power, keeping or shifting agency or attributing it to others, or masking between the positions as they tell the story [50].

Before starting the actual data collection, a small pilot study will take place, following the advice of [48]. This is important to rehearse and validate the chosen approach and instruments. To vet the instruments the dialogic sessions with our peers will be used, together with conversations with other brilliant qualitative researchers in the field from our network and, hopefully, - at CHI itself. Their feedback will be integrated into the selected research process iteratively. The next step would be a rehearsal interview with one of our peers, followed up by a debriefing session with them. The reflections and insights will shape and refine the research protocols even further.

Interviews will take place online via Zoom, invitations for the call will be sent to the participants in advance together with the link where they can join and tech instructions/criteria. On the side of the interviewer, it will be a quiet, controlled, private environment that will be disclosed to the participant beforehand. Participants will be encouraged in their invitation letters to select not only a quiet and private environment for themselves but also a space where they feel comfortable, safe, and relaxed. They will also be encouraged to prepare small snacks and a sufficient amount of water. Allowing participants to choose the setting for themselves is one of the test validity constraints described by [45].

3.2 Data analysis and interpretation

Narratives do not speak for themselves; they require interpretation and analysis. We will approach the data by making sure that we keep the fidelity to the participants, always keeping in mind that they are the owners and the experts of their own stories. We will recognize our authority, as interpreters, during each of the steps, and will reflectively critique our positionality and the assumptions it brings to the process. Data analysis will be a generative and recursive process; the data gathered during the study will consist of the transcripts of the recorded interviews, and our research journal with memos, notes, thoughts, reflections, and critiques. We will begin by reviewing them holistically and identify segments of the most relevance and significance to the study. After the transcripts are done, a thematic analysis strategy will be used to extract themes that can be noticed across the stories. Thematic analysis gives the narratives sense and direction [22]. “Divergent” stories or contradicting experiences will be presented as such. Each participant will

be given the opportunity to check if the analysis and analyst interpretation and description still fits their experience. In cases where deviation or mistranslation has occurred, analyst and interviewee will dialogue to clarify and changes will be reflected in the final presentation of themes.

Transcripts will be typed, printed, and coding will be tracked using a reflection journal by the analyst. Coding will occur in two passes approximately one month apart to allow for some over-determination of themes. The first pass will preserve the temporal nature of expressions and note emerging themes. The second pass will be a closer read, segmenting expressions and ascribe each segment to themes. Themes from individual narratives will be cross-compared with other transcriptions and grouped. After this process is completed the analyst will validate themes and organisation from the second pass, fitting them to the linear order of the first – essentially restorying from a grouped perspective that, at times, will diverge where interviewee experiences have diverged [9].

3.3 Quality criteria

Ravitch and Mittenfelner [48] describe five quality criteria. Credibility is reached by analysing the complexity of reality and applying the intersectionality lens, to investigate the unique variations of experiences. The gathered insights can be applied to other parts of the educational systems, the same biases most likely will be present not only in gamification – this creates transferability. The study is reliable and we have shown above how the selected approaches will help to answer the research questions. To provide confirmability, we made sure to take into account our own biases as much as possible, by constant self-reflection and checking with our positioned experiences and known biases. Participants will be invited to co-create the data gleaned from their experiences, stepping in across multiple stages of analysis and composition—especially where suggestions for best practices and methods for correcting discriminatory gamified tools are concerned.

3.4 Validity

Strategies to increase validity have been introduced, from incorporation of constructionist and intersectional frameworks through data collection to interview strategies. Structured reflexivity is incorporated from the author's situatedness to allow participant narratives to tell themselves with little analysis. Allowing participants to review the data from analysis as written in the study improves interpretive validity while the diversity of perspectives presented facilitates descriptive and generalizable validity. Analysis has several validation steps, internal and external, as described above. As this study's purpose is more exploratory and diversity driven than generalising and universalising, validity is primarily concerned with participants' feeling their story has been told and finding moments of convergence and divergence between others' interviewee experiences.

4 IMPLICATIONS FOR THE CHI COMMUNITY

As Gray [20] discusses, there is value in determining whether our technology is reproducing analogue biases, creating new forms of bias and discrimination, or being used inclusively. Social norms within education are slow to change, and while it may be the case

that innovations in gamified education technology are helping reduce bias in the classroom, case studies from artificial intelligence, digital games, and virtual reality suggest otherwise [20]. While gamified education technologies permit an incredible amount of creative freedom in how knowledge is disseminated and skills are honed, understanding how these biases feature in the stories of marginalised people is crucial to correcting these issues before they become normalised, suppressed, and harder to adjust. We are certain that this type of knowledge is crucial for the researchers in the field of HCI, and we're hoping to provoke more research and discussion on this subject.

5 ANTICIPATED RESULTS AND CONCLUSION

The main anticipated result of the whole project is a deeper understanding of biases and marginalising experiences while using existing gamified tools. As normalisation of recurring discriminatory stressors can mask or suppress these experiences [40], rigorous and careful exploratory work of discriminatory stressors is necessary. The narrative approach should help to engage the participants in storytelling about their experiences in detail. The stories collected should help to unravel which biases are present from which tools and their effects on students' experiences and perceptions. Both outcomes are important precedents for expanding our critical and liberatory practices around gamified education tools as well as any future studies seeking to accurately quantify the impacts of these biases. The concluding result of this paper in particular, is refining our research questions and situating them in the existing body of literature. We hope to collaborate with the HCI community for the next steps and to provoke more research about the implicit biases in gamification.

To situate this project with the broader research field, we hope to be able to discover the relations between these biases and the ones experienced in non-gamified educational systems. This will also help to extend our understanding of the "bottom-up" gamification, described in [29], when we allow users to pick the gamified elements they find the most relative. Having all these questions answered will contribute to the existing research by providing more in-depth insights into the already existing quantitative data on the usage of gamification in education, and specifically will expand our understanding of its implicit effects.

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A IN DEPTH INTERVIEW QUESTIONS (45 MINUTES)

- (1) How often do you play games, if at all? Do games have a role in your self-development or learning strategies?
- (2) Are you familiar with the concept of gamification and the difference between gamification and educational games?
 - If yes - what does “gamification” mean to you?
 - If not - what do you think it might or could be?
- (3) Can you tell me about some of your experiences with gamification in educational settings?
 - Which gamification tools have you encountered in educational contexts? When? Describe your experiences.
 - Can you describe elements of the gamified tools that stood out?
 - Did they work as intended? Which parts did, which didn't?
 - Did you feel a personal relationship with the following: avatars, settings, iconography, content, badges, story, etc.
 - How do you feel this experience would have gone if these elements were more closely related to who you are? What would have to change? What would you have done differently if you were able to develop this gamified tool for your younger self?
 - Have there been experiences where who you are was not present in gamified tools? If so, describe your experiences. What effect did that have on you in the moment? If any, can you describe some of the long term effects?
- (4) Can you tell me some of your experiences with education?
 - Can you describe how your identity has been accommodated in educational settings?
 - Can you recall any empowering experiences? Please, describe
 - Can you recall any disempowering experiences? Please, describe
- (5) When you think about how your identity has featured in the education you've received (e.g. the content, how it was delivered, classmates, mentors, connections, tools, technology, etc.), what would you change about the classroom setting for individuals like yourself, if anything? Why?
 - Do you think gamification can play a role in these changes? Why/not?
 - In an ideal world, would your perfect educational setting have gamification elements in them?
 - If so, what would those games look like? What elements would they have/not have?
 - If not, why?